



Indian Institute of Information Technology, Surat
Department of Electronics & Communication Engineering
End Semester Examination, 4th Sem, 2024-25
Subject Name: Analog & Digital Communication (EC-407)

Timing: 10:00 AM to 01:00 PM

Date: 14/05/2025

Max Marks: 100

- Note:**
1. All questions are compulsory.
 2. Assume suitable data wherever necessary.
 3. Draw suitable diagrams when required
 4. Mention the Question number clearly.

1.	(a)	Find the Fourier transform of the given signal and plot the time domain and frequency spectrum for the same. $x(t) = \begin{cases} \frac{1}{2} & \frac{1}{2} \leq t < 1 \\ 1 & t \leq \frac{1}{2} \end{cases}$	(8)
	(b)	With proper examples to justify the statement, the total area under the curve of the power spectral density of a power signal is equal to the average power of that signal.	(4)
	(c)	Obtain DFT of delayed unit impulse $\delta(n - n_0)$.	(6)
	(d)	Compute the IDFT of four points sequence $X(k) = \{4, 1 - j, -2, 1 + j\}$	(2)
2.	(a)	Explain in detail about different types of modulation based on the modulation index of AM with a diagram.	(3)
	(b)	An antenna has an impedance of 40 Ω . An unmodulated carrier signal produces a current of 4.8 A. The modulation is 90 percent. Calculate (a) the carrier power, (b) The total power, and (c) The sideband power.	(3)
	(c)	Explain in detail about envelope detector of AM signal with their disadvantages.	(4)
	(d)	Explain in detail about the phase discrimination method of SSB-SC signal.	(3)
3.	(a)	An FM signal is generated by modulating a carrier of frequency $f_c = 100$ MHz with a sinusoidal message signal of frequency $f_m = 15$ kHz and amplitude $A_m = 3$ V. The frequency sensitivity of the modulator is $k_f = 20$ kHz/V. (i) Calculate the modulation index β . (ii) Using Bessel function values, determine the significant spectral components and their relative amplitudes up to the third order. (iii) Estimate the total power in the carrier and first three sidebands (up to $n=3$). (iv) Estimate the bandwidth based on the significant sidebands only (not Carson's Rule).	(10)
	(b)	Describe the working of a Phase-Locked Loop (PLL) FM demodulator.	(5)
	(c)	A continuous-time analog signal is bandlimited to 4 kHz and is sampled at a rate of 10 kHz. The signal is then quantized using a uniform quantizer with 8 levels, and encoded using standard Pulse Code Modulation (PCM). (a) What is the minimum sampling rate required according to the Nyquist theorem? Is the chosen sampling rate sufficient? (b) How many bits per sample are required to represent the quantized signal? (c) Calculate the bit rate of the PCM signal. (d) Explain briefly the significance of quantization noise in PCM and how it can be reduced.	(10)

4.	(a)	Explain the working principle of a superheterodyne receiver with a neat block diagram.	(6)
	(b)	In a superheterodyne receiver, the received RF signal is at 900 kHz, and the intermediate frequency is 455 kHz. - Determine the local oscillator frequency. - Calculate the image frequency. - Explain why the RF amplifier stage is important for rejecting the image frequency.	(5)
	(c)	Explain the principle of Automatic Gain Control (AGC) in communication systems.	(5)
5.	(a)	Explain the concept of the T Carrier System. The T1 line is used for transmitting digital signals with a data rate of 1.544 Mbps. The system transmits data in 24 channels, each with a bandwidth of 64 kbps. (i) Calculate the total data rate of the T1 system. (ii) In a T1 system, each channel can carry voice signals. Calculate the maximum number of voice calls that can be carried simultaneously over a T1 line, assuming that each voice call requires 64 kbps.	(8)
	(b)	Draw a block diagram of a TDM system for transmitting 4 user signals in detail and label the frame structure.	(6)
6.	(a)	Draw a signal waveform for the given data sequence using unipolar, Manchester, NRZ-I, NRZ-L and Bipolar Encoding. Binary data sequence: 10110101.	(6)
	(b)	A communication channel has a bandwidth of 20 kHz, and the signal-to-noise ratio (S/N) is 30 dB. Calculate: (i) The Shannon capacity of the channel. (ii) The minimum number of discrete signal levels N required to achieve the data rate calculated in (i), assuming Nyquist criterion is used.	(6)

****All the Best****